SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

Chennai- 602105

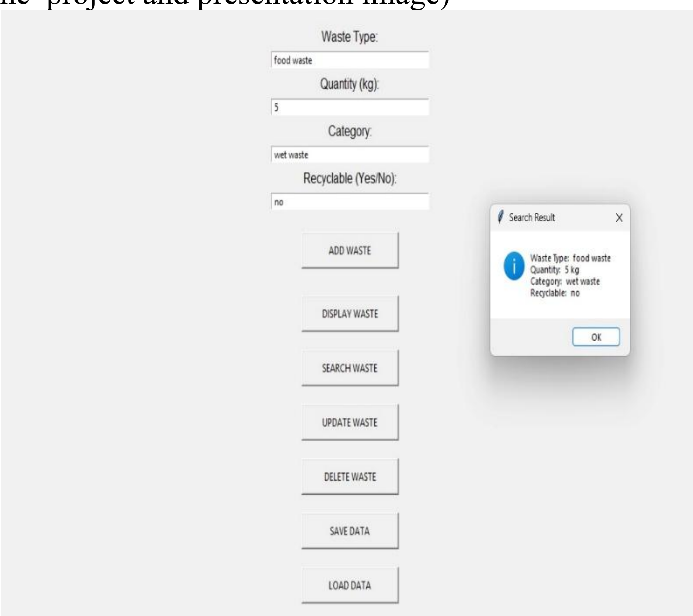
Student Name: P.V. KARISHMIKA **Reg.No.:** 192511453

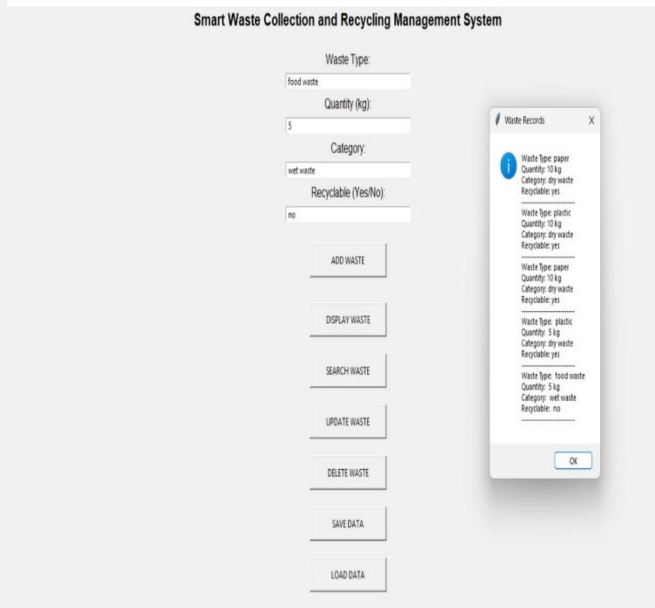
Course Code: CSA0802 **Slot:** C

Course Name: PYTHON

Course Faculty: Dr. A.Jegatheesan

Project Title: WASTE MANAGEMENT SYSTEM
Module: DICTIONARY OPERATIONS
Module Photographs : (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)





Project Description : (here you write what you did in this project (contribution) including Model Description

Individual Contribution – Module 1: Dictionary Operations

The first module of the **Smart Waste Collection and Recycling Management System** focuses on managing waste information using Python dictionaries. The main objective of this module is to provide an efficient way to store, access, modify, and remove waste records. I contributed to the design and implementation of the dictionary-based operations used in the project.

I implemented the waste data dictionary to store information about different types of waste. The waste type is used as the dictionary key, while the quantity, category, and recyclable status are stored as values inside another dictionary. This structure allows multiple details of each waste type to be stored in an organized manner.

I worked on the **Add Waste** operation, where the user enters the waste type, quantity, category, and recyclable status through the Tkinter input fields. The entered information is then stored in the dictionary as a key-value pair. I also implemented input validation to ensure that the waste type is entered before adding a record.

I contributed to the **Display Waste** operation by using the items () method of the dictionary. This operation accesses every waste type and its corresponding details and displays them to the user. The **Search Waste** operation was implemented using the in operator to check whether a particular waste type exists in the dictionary. If the record exists, its details are retrieved and displayed; otherwise, an appropriate error message is shown.

I also worked on the **Update Waste** operation. When an existing waste type is entered, its quantity, category, and recyclable status can be replaced with new values. The **Delete Waste** operation uses the del statement to remove an existing waste record from the dictionary.

Testing and debugging were an important part of my contribution to this module. I tested different waste types such as Plastic, Paper, and Food Waste and verified that adding, displaying, searching, updating, and deleting records worked correctly. I also checked error conditions such as empty input fields and searching for records that do not exist.

Through this module, I gained practical knowledge of Python dictionaries, key-value pairs, nested dictionaries, dictionary methods, conditional statements, loops, functions, input validation, and debugging. This module forms the main data management component of the waste management system.

Student Signature **Guide Signature**